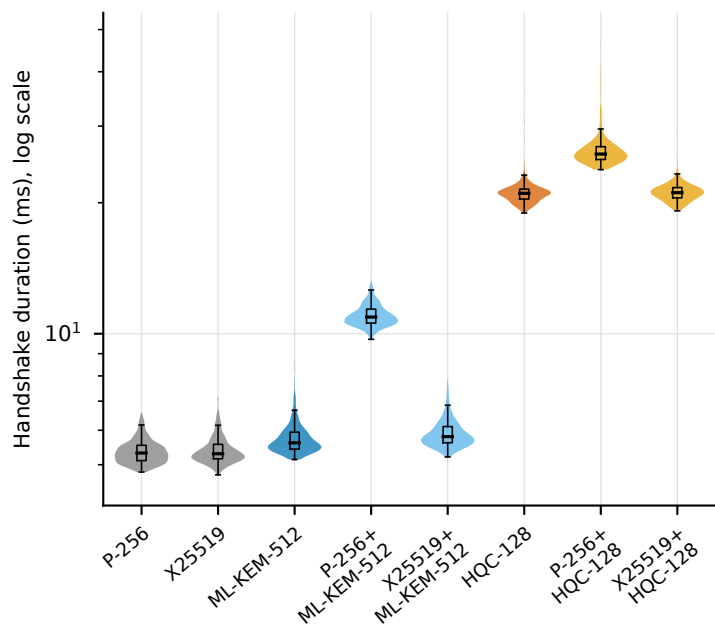


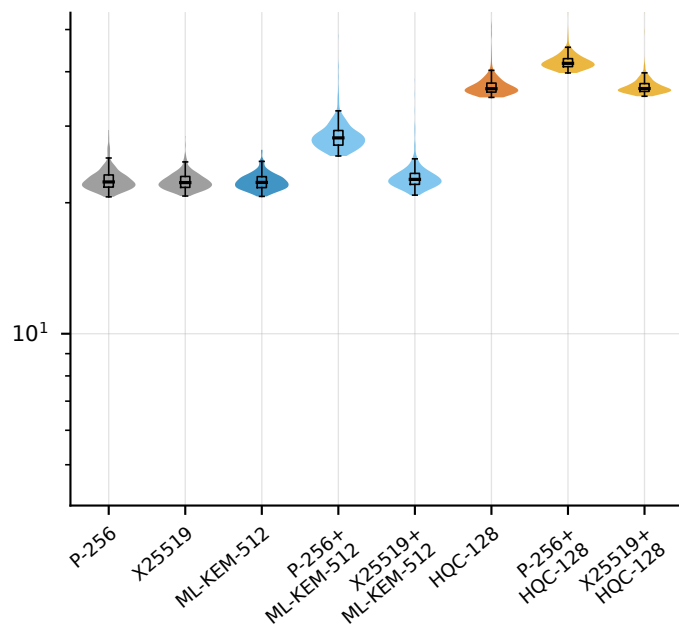
(a)

ed25519 (classical)



(b)

mlds44 (post-quantum)



(c)

 Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	mlds44 Mean (ms)	mlds44 CV (%)	mlds44 Max (ms)
P-256	5.38	6.4	6.58	22.6	5.9	29.4
X25519	5.42	7.1	7.15	22.4	4.6	28.5
ML-KEM-512	5.75	8.0	8.69	22.4	4.4	26.5
P-256+ML-KEM-512	11.0	6.0	15.3	28.6	8.0	48.7
X25519+ML-KEM-512	5.92	7.5	7.86	22.9	7.0	38.6
HQC-128	21.0	5.8	36.2	37.2	7.0	58.6
P-256+HQC-128	26.4	8.6	45.9	42.5	7.3	71.0
X25519+HQC-128	21.2	5.3	31.7	37.0	4.3	55.2